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Semiconductor device with improved cell density and manufacturing method of the semiconductor device

Abstract

A semiconductor device includes a gate structure including conductive layers and insulating layers alternately stacked with each other, channel structures passing through the gate structure and arranged in a first direction, a cutting structure extending in the first direction and passing through the channel structures, and a first slit structure passing through the gate structure and extending in a second direction crossing the first direction.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8748971	12/2013	Tanaka et al.	N/A	N/A
2017/0077108	12/2016	Kawaguchi	N/A	H10B 43/27
2018/0277554	12/2017	Kaneko	N/A	H01L 21/7682
2020/0343307	12/2019	Lee	N/A	H10B 63/34

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107293549	12/2016	CN	N/A
108231781	12/2017	CN	N/A
109801922	12/2018	CN	N/A
112242402	12/2020	CN	N/A
1020160008404	12/2015	KR	N/A
1020160016430	12/2015	KR	N/A
1020180073161	12/2017	KR	N/A
1020200125148	12/2019	KR	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) The present application claims priority under 35 U.S.C. § 119(a) to Korean patent application number 10-2021-0053233, filed on Apr. 23, 2021, which is incorporated herein by reference in its entirety.
BACKGROUND
1. Technical Field
(2) The disclosed technology generally relates to an electronic device, and more particularly, to a semiconductor device and a method of manufacturing the semiconductor device.
2. RELATED ART

(3) The integration density of a semiconductor device may be determined mainly by an area of a unit memory cell. The increase in integration density of a semiconductor device in which memory cells are formed in a single layer over a substrate has recently been limited. Thus, a three-dimensional semiconductor device has been proposed in which memory cells are stacked over a substrate. In addition, to improve the operational reliability of semiconductor devices, various structures and manufacturing methods have been developed.

SUMMARY
(4) Various embodiments are directed to a semiconductor device having a stable structure and improved characteristics, and a method of manufacturing the semiconductor device.
(5) According to an embodiment, a semiconductor device may include a gate structure including conductive layers and insulating layers alternately stacked with each other, channel structures passing through the gate structure and arranged in a first direction, a cutting structure extending in the first direction and passing through the channel structures, and a first slit structure passing through the gate structure and extending in a second direction crossing the first direction.
(6) According to an embodiment, a semiconductor device may include a gate structure including conductive layers and insulating layers alternately stacked with each other, pillar structures passing through the gate structure, a cutting structure passing through the pillar structures and separating each of the pillar structures into a first pillar structure and a second pillar structure, a first slit structure passing through the gate structure and extending in a direction crossing the cutting structure, a first interconnection line extending in a direction crossing the first slit structure and coupled to the first pillar structures, and a second interconnection line extending in a direction crossing the first slit structure and coupled to the second pillar structures.
(7) According to an embodiment, a method of manufacturing a semiconductor device may include forming a stacked structure, forming channel structures passing through the stacked structure and arranged in a first direction, forming a cutting structure passing through the channel structures and extending in the first direction, and forming a first slit structure passing through the stacked structure and extending in a second direction crossing the first direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIGS. **1**A to **1**D are diagrams illustrating the structure of a semiconductor device according to an embodiment of the present disclosure;
(2) FIGS. **2**A to **2**C are diagrams illustrating the structure of a semiconductor device according to an embodiment of the present disclosure;
(3) FIGS. **3**A to **3**C are diagrams illustrating the structure of a semiconductor device according to an embodiment of the present disclosure;
(4) FIGS. **4**A to **4**D are diagrams illustrating the structure of a semiconductor device according to an embodiment of the present disclosure;
(5) FIGS. **5**A and **53**, FIGS. **6**A and **63**, FIGS. **7**A to **7**C, and FIGS. **8**A to **8**C are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure;
(6) FIGS. **9**A and **93** are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure;
(7) FIG. **10** is a diagram illustrating a memory system according to an embodiment of the present disclosure;
(8) FIG. **11** is a diagram illustrating a memory system according to an embodiment of the present disclosure;
(9) FIG. **12** is a diagram illustrating a memory system according to an embodiment of the present

disclosure;
(10) FIG. **13** is a diagram illustrating a memory system according to an embodiment of the present disclosure; and
(11) FIG. **14** is a diagram illustrating a memory system according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(12) Specific structural or functional descriptions of examples of embodiments in accordance with concepts which are disclosed in this specification are illustrated only to describe the examples of embodiments in accordance with the concepts and the examples of embodiments in accordance with the concepts may be carried out by various forms but the descriptions are not limited to the examples of embodiments described in this specification.
(13) FIGS. **1**A to **1**D are diagrams illustrating the structure of a semiconductor device according to an embodiment of the present disclosure.
(14) Referring to FIGS. **1**A to **1**C, a semiconductor device may include a gate structure GST, pillar structures P, a cutting structure CS, and a first slit structure SLS**1**, The semiconductor device may include a base **10**, a second slit structure SLS**2**, a first contact plug CT**1**, a second contact plug CT**2**, or a combination thereof.
(15) The gate structure GST may include conductive layers **11** and insulating layers **12** that are stacked alternately with each other. Each of the conductive layers **11** may be a gate electrode of a memory cell or a select transistor. The conductive layers **11** may include a conductive material such as polysilicon or metal (e.g., tungsten, molybdenum). The insulating layers **12** may insulate the stacked conductive layers **11** from each other. The insulating layers **12** may include an insulating material such as oxides, nitrides, or air gaps.
(16) The gate structure GST may be located on the base **10**. The base **10** may be a semiconductor substrate or a source layer. The semiconductor substrate may include a source region which is doped with impurities. The source layer may include a conductive material such as polysilicon, tungsten, molybdenum, or metal.
(17) The pillar structures P may pass through the gate structure GST. The pillar structures P may be arranged in a first direction I and a second direction II crossing the first direction I, Crossing directions means the directions are not parallel. For example, the directions may be substantially perpendicular to one another. According to an embodiment, the pillar structures P may be arranged in a matrix format.
(18) Each of the pillar structures P may include a pair of a first pillar structure P**1** and a second pillar structure P**2**. Each of the pillar structures P may be split into one pair of the first pillar structure P**1** and the second pillar structure P**2** by the cutting structure CS. The one pair of the first pillar structure P**1** and the second pillar structure P**2** may neighbor each other in the second direction II with the cutting structure CS interposed therebetween, or may have a symmetrical structure with respect to the cutting structure CS.
(19) According to an embodiment, each of the pillar structures P may be a channel structure which includes channel layers **13**A and **13**B. The first pillar structure P**1** may be a first channel structure and the second pillar structure P**2** may be a second channel structure. First memory cells or select transistors may be arranged at intersections of the first pillar structure P**1** and the conductive layers **11**, Second memory cells or select transistors may be arranged at intersections of the second pillar structure P**2** and the conductive layers **11**. The first memory cell and the second memory cell that neighbor each other in the second direction II with the cutting structure CS interposed therebetween may be driven independently of each other.
(20) The first pillar structure P**1** may include a first channel layer **13**A. The first channel layer **13**A may refer to a region where a channel of a memory cell, a select transistor, and the like is formed. The first channel layer **13**A may include a semiconductor material such as silicon or germanium. The first pillar structure P**1** may further include a first conductive pad **14**A. The first conductive

pad **14**A may be coupled to the first channel layer **13**A and include a conductive material. The first pillar structure P**1** may include a first insulating core **15**A. The first insulating core **15**A may include an insulating material such as oxides, nitrides, and air gaps. The first pillar structure P**1** may further include a memory layer (not illustrated) which is located between the first channel layer **13**A and the conductive layers **11**. The memory layer may include at least one of a tunneling layer, a data storage layer, and a blocking layer. The data storage layer may include a floating gate, a charge trapping material, polysilicon, a nitride, a variable resistance material, a nanostructure, or a combination thereof.

(21) The second pillar structure P**2** may have a similar structure to the first pillar structure P**1**. The second pillar structure P**2** may include the second channel layer **13**B, The second pillar structure P**2** may further include a second conductive pad **14**B, a second insulating core **15**B, or a combination thereof.

(22) According to an embodiment, each of the pillar structures P may be an electrode structure that includes an electrode layer. The first pillar structure P**1** may be a first electrode structure and the second pillar structure P**2** may be a second electrode structure. The first electrode structure may include a first electrode layer instead of the first channel layer **13**A. The first electrode structure may further include a first conductive pad **14**A, a first insulating core **15**A, or a combination thereof. The first pillar structure P**1** may further include a memory layer (not illustrated) which is located between the first electrode layer and the conductive layers **11**. The second electrode structure may include a second electrode layer instead of the second channel layer **13**B. The second electrode structure may further include the second conductive pad **14**B, the second insulating core **15**B, or a combination thereof. The second pillar structure P**2** may further include a memory layer (not illustrated) which is located between the second electrode layer and the conductive layers **11**.

(23) The cutting structure CS may pass through the pillar structures P and extend to the base **10**. The cutting structure CS may pass through the gate structure GST and the pillar structures P**1** and extend in the first direction I. The cutting structure CS may successively pass through the pillar structures P. The cutting structure CS may cross at least two of the pillar structures P arranged in the first direction I and may separate each pillar structure P into a pair of the first pillar structure P**1** and the second pillar structure P**2**. The cutting structure CS may include an insulating material such as oxides, nitrides, and air gaps.

(24) A plurality of cutting structures CS may be located between one pair of the first slit structures SLS**1**. The cutting structures CS may be arranged in the first direction I and the second direction II. According to an embodiment, the cutting structures CS may be arranged in a matrix format.

(25) The first slit structure SLS**1** may pass through the gate structure GST. The first slit structure SLS**1** may extend in a direction crossing the cutting structure CS. The first slit structure SLS**1** may extend in the second direction According to an embodiment, the first slit structure SLS**1** may be arranged perpendicular to the cutting structure CS. The first slit structure SLS**1** may include an insulating material. According to an embodiment, the first slit structure SLS**1** may include a contact structure which is electrically coupled to the base **10** and an insulating spacer which insulates the contact structure and the conductive layers **11** from each other.

(26) The second slit structure SLS**2** may pass through the gate structure GST at a shallower depth than the first slit structure SLS**1** or the cutting structure CS. The second slit structure SLS**2** may have a depth that passes through at least one uppermost conductive layer **11**. According to an embodiment, the second slit structure SLS**2** may have a depth such that the second slit structure SLS**2** passes through at least one conductive layer **11** corresponding to a select line among the conductive layers **11**, and does not pass through the conductive layers **11** corresponding to word lines.

(27) At least one second slit structure SLS**2** may be located between one pair of the first slit structures SLS**1**. The second slit structure SLS**2** may extend in a direction crossing the cutting structure CS, The second slit structure SLS**2** may extend in parallel with the first slit structure

SLS**1**. The second slit structure SLS**2** may extend in the second direction II. The cutting structures CS may be arranged symmetrically or asymmetrically at both sides with respect to the second slit structure SLS**2**. The second slit structure SLS**2** may contact at least one cutting structure CS, The pillar structures P may be located between the first slit structure SLS**1** and the second slit structure SLS**2**. Some of the pillar structures P may contact the second slit structure SLS**2**.
(28) Referring to FIGS. **1**A and **1**D, the semiconductor device may further include a first interconnection line IL**1** and a second interconnection line IL**2**, The first interconnection line IL**1** and the second interconnection line IL**2** may extend in a direction crossing the first slit structure SLS**1** or the second slit structure SLS**2**. The first interconnection line IL**1** and the second interconnection line IL**2** may run parallel with the cutting structure CS and extend in the first direction I.
(29) According to an embodiment, the first contact plug CT**1** may be coupled to a first pillar structure P**12** and the second contact plug CT**2** may be coupled to the first contact plug CT**1**. According to an embodiment, the first contact plug CT**1** and the second contact plug CT**2** may be located at different levels, and an upper surface of the first contact plug CT**1** and a bottom surface of the second contact plug CT**2** may be coupled to each other. A first interconnection line IL**11** may be coupled to a first pillar structure P**12** through the first contact plug CT**1** and the second contact plug CT**2**. A first interconnection line IL**12** may be coupled to a first pillar structure P**11** through the first contact plug CT**1** and the second contact plug CT**2**. A second interconnection line IL**21** may be coupled to a second pillar structure P**22** through the first contact plug CT**1** and the second contact plug CT**2**. A second interconnection line IL**22** may be coupled to a second pillar structure P**21** through the first contact plug CT**1** and the second contact plug CT**2**.
(30) However, the numbers of the first slit structures SLS**1**, the second slit structures SLS**2**, and the pillar structures P as shown in FIGS. **1**A and **1**D may vary. For example, the number of pillar structures P located between one pair of the first slit structures SLS**1**, the number of pillar structures P located between the first slit structure SLS**1** and the second slit structure SLS**2**, the number of cutting structures CS located between on pair of the first slit structures SLS**1**, and the number of cutting structures CS located between the first slit structure SLS**1** and the second slit structure SLS**2** may vary.
(31) According to the above-described structure, one pillar structure P may be separated into a plurality of pillar structures P**1** and P**2** using the cutting structure CS. Therefore, the number of memory cells realized with one pillar structure P may be increased. Therefore, the number of memory cells included in the gate structure GST may be increased even when the number of stacked conductive layers **11** included in the gate structure GST is not increased.
(32) FIGS. **2**A to **2**C are diagrams illustrating the structure of a semiconductor device according to an embodiment of the present disclosure. Hereinafter, repeated descriptions of certain elements that are described above are omitted for brevity.
(33) Referring to FIGS. **2**A to **2**C, a semiconductor device may include the gate structure GST, the pillar structures P, the cutting structure CS, and the first slit structure SLS**1**. The semiconductor device may include the base **10**, the second slit structure SLS**2**, the first contact plug CT**1**, the second contact plug CT**2**, or a combination thereof.
(34) The pillar structures P may be staggered with respect to each other. According to an embodiment, centers of the pillar structures P which are adjacent to each other in the first direction I may coincide with each other, while centers of the pillar structures P which are adjacent to each other in the second direction II may be offset from each other.
(35) The cutting structures CS may be staggered with respect to each other. According to an embodiment, centers of the cutting structures CS which are adjacent to each other in the first direction I may coincide with each other, while centers of the cutting structures CS which are adjacent to each other in the second direction II may be offset from each other.
(36) Two or more second slit structures SLS**2** may be located between one pair of the first slit

structures SLS**1**. FIG. **2**A, for example, shows two second slit structures SLS**2** between a pair of first slit structures SLS**1**. The pillar structures P may be located between the first slit structure SLS**1** and the second slit structure SLS**2** and between the second slit structures SLS**2**.

(37) The second slit structure SLS**2** may contact some of the cutting structures CS at both sides thereof. The second slit structure SLS**2** may contact the cutting structure CS at one side thereof and be separated from the cutting structure CS at the other side thereof.

(38) FIGS. **3**A to **3**C are diagrams illustrating the structure of a semiconductor device according to an embodiment of the present disclosure. Hereinafter, repeated descriptions of certain elements that are described above are omitted for brevity.

(39) Referring to FIGS. **3**A and **3**B, a semiconductor device may include the gate structure GST, the pillar structures P, the cutting structure CS, and the first slit structure SLS**1**. The semiconductor device may further include the base **10**, the second slit structure SLS**2**, or a combination thereof.

(40) The cutting structure CS may pass through three or more pillar structures P that are arranged in the first direction I. The second slit structure SLS**2** may have a zigzag shape, such as a shape including non-parallel line segments joined end-to-end, or a wave shape, such as a shape including curves joined end-to-end. A zigzag shape, for example, is pictured in FIG. **3**A. The second slit structure SLS**2** may be separated from the cutting structures CS at both sides thereof.

(41) Referring to FIG. **3**C, the semiconductor device may further include the first interconnection line IL**1** and the second interconnection line IL**2**. The first interconnection line IL**1** and the second interconnection line IL**2** may extend in the first direction I.

(42) According to an embodiment, the first interconnection line IL**11** may be coupled to a first pillar structure P**13** through the first contact plug CT**1** and the second contact plug CT**2**, The first interconnection line IL**12** may be coupled to the first pillar structure P**12** through the first contact plug CT**1** and the second contact plug CT**2**. A first interconnection line IL**13** may be coupled to the first pillar structure P**11** through the first contact plug CT**1** and the second contact plug CT**2**. The second interconnection line IL**21** may be coupled to a second pillar structure P**23**. The second interconnection line IL**22** may be coupled to the second pillar structure P**22**. A second interconnection line IL**23** may be coupled to the second pillar structure P**21**.

(43) FIGS. **4**A to **4**D are diagrams illustrating the structure of a semiconductor device according to an embodiment of the present disclosure. Hereinafter, repeated descriptions of certain elements that are described above are omitted for brevity.

(44) Referring to FIGS. **4**A to **4**C, a semiconductor device may include the gate structure GST, the pillar structures P, the cutting structure CS, and the first slit structure SLS**1**. The semiconductor device may include the base **10**, the second slit structure SLS**2**, the first contact plug CT**1**, the second contact plug CT**2**, or a combination thereof.

(45) Each of the pillar structures P may include the first pillar structure P**1** and the second pillar structure P**2**. The first pillar structure P**1** may include a first sub pillar structure PIA and a second sub pillar structure P**1**B. The second pillar structure P**2** may include a first sub pillar structure P**2**A and a second sub pillar structure P**2**B.

(46) The second sub pillar structures P**1**B and P**2**B may include at least one uppermost conductive layer among the conductive layers **11**. The first sub pillar structures PIA and P**2**A may pass through the remaining conductive layers **11** among the conductive layers **11**, According to an embodiment, the first sub pillar structures P**1**A and P**2**A may correspond to a memory cell or a source select transistor, and the second sub pillar structures P**1**B and P**2**B may correspond to a drain select transistor.

(47) The first sub pillar structure PIA may include the first channel layer **13**A, the first conductive pad **14**A, the first insulating core **15**A, or a combination thereof. The second sub pillar structure P**1**B may include a first channel layer **23**A, a first conductive pad **24**A, a first insulating core **25**A, or a combination thereof. The first sub pillar structure P**2**A may include the second channel layer **13**B, the second conductive pad **14**B, the second insulating core **15**B, or a combination thereof. The

second sub pillar structure P2B may include a second channel layer **23**B, a second conductive pad **24**B, a second insulating core **25**B, or a combination thereof. A first electrode layer and a second electrode layer may replace the first channel layer **13**A or **23**A and the second channel layer **13**B or **23**B.

(48) Referring to FIG. **4**D, the semiconductor device may further include the first interconnection line IL**1** and the second interconnection line IL**2**. The first interconnection line IL**1** and the second interconnection line IL**2** may extend in the first direction I.

(49) According to an embodiment, the first pillar structure P**11** may include a first sub pillar structure P**11**A and a second sub pillar structure PUB, The first pillar structure P**12** may include a first sub pillar structure P**12**A and a second sub pillar structure P**12**B. The first pillar structure P**13** may include a first sub pillar structure P**13**A and a second sub pillar structure P**13**B. A first pillar structure P**14** may include a first sub pillar structure P**14**A and a second sub pillar structure P**14**B. The first contact plug CT**1** may be coupled to the second sub pillar structures PUB to P**14**B of the first pillar structures P**11** to P**14**. The second contact plug CT**2** may be coupled to the first contact plug CT**1**.

(50) The second pillar structure P**21** may include a first sub pillar structure P**21**A and a second sub pillar structure P**21**B. The second pillar structure P**22** may include a first sub pillar structure P**22**A and a second sub pillar structure P**22**B. The second pillar structure P**23** may include a first sub pillar structure P**23**A and a second sub pillar structure P**23**B, A second pillar structure P**24** may include a first sub pillar structure P**24**A and a second sub pillar structure P**24**B. The first contact plug CT**1** may be coupled to the second sub pillar structures P**21**B to P**24**B of the second pillar structures P**21** to P**24**. The second contact plug CT**2** may be coupled to the first contact plug CT**1**. The first interconnection line IL**11** may be coupled to the first pillar structure P**12** and the first pillar structure P**14** through the first contact plugs CT**1** and the second contact plugs CT**2**, The first interconnection line IL**12** may be coupled to the first pillar structure P**11** and the first pillar structure P**13** through the first contact plugs CT**1** and the second contact plugs CT**2**. The second interconnection line IL**21** may be coupled to the second pillar structure P**22** and the second pillar structure P**24** through the first contact plugs CT**1** and the second contact plugs CT**2**. The second interconnection line IL**22** may be coupled to the second pillar structure P**21** and the second pillar structure P**23** through the first contact plugs CT**1** and the second contact plugs CT**2**.

(51) FIGS. **5**A and **5**B, FIGS. **6**A and **6**B, FIGS. **7**A to **7**C, and FIGS. **8**A to **8**C are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure.

(52) Referring to FIGS. **5**A and **5**B, a stacked structure ST may be formed on a base **50**. The base **50** may be a semiconductor substrate, a source structure, or the like. The semiconductor substrate may include a source region doped with impurities. The source structure may include a source layer including a conductive material such as polysilicon, tungsten, molybdenum, or metal. Alternatively, however, the source region may include a sacrificial layer to be replaced with a source layer during subsequent processes.

(53) The stacked structure ST may be formed by alternately forming first material layers **51** and second material layers **52**. The first material layers **51** may include a material having a high etch selectivity with respect to the second material layers **52**. For example, the first material layers **51** may include a sacrificial material, such as a nitride, and the second material layers **52** may include an insulating material, such as an oxide. In another example, the first material layers may include a conductive material, such as polysilicon, tungsten, or molybdenum, and the second material layers **52** may include an insulating material, such as an oxide.

(54) Subsequently, the pillar structures P may be formed through the stacked structure ST. The pillar structures P may be arranged in the first direction I and the second direction II crossing the first direction I, The pillar structures P neighboring each other in the first direction I may be arranged so that centers thereof may coincide with each other. On the other hand, the pillar

structures P neighboring each other in the second direction II may be arranged so that centers thereof may be offset from each other.
(55) In the plane defined by the first direction I and the second direction II, the pillar structure P may have various shapes such as a circle, an ellipse, and a polygon. A planar cross section of the pillar structure P may have a first width W**1** in the first direction I and a second width W**2** in the second direction II. The first width W**1** and the second width W**2** may be the same as or different from each other. In consideration of the width of a cutting structure to be formed during subsequent processes, the second width W**2** may be greater than the first width W**1**.
(56) Each of the pillar structures P may include a channel layer **53**. According to an embodiment, after an opening is formed through the stacked structure ST, the channel layer **53** may be formed in the opening. A memory layer may be formed before the channel layer **53** is formed. After an insulating core **55** is formed, a conductive pad **54** may be formed, Each of the pillar structures P may include an electrode layer instead of the channel layer **53**. The insulating core **55** or the conductive pad **54** may be omitted.
(57) Referring to FIGS. **6**A and **6**B, cutting structures **56** may be formed. Each of the cutting structures **56** may pass through at least two pillar structures P and extend in the first direction I. Each of the pillar structures P may be separated into the first pillar structure P**1** and the second pillar structure P**2**.
(58) The first pillar structure P**1** may be a first channel structure and the second pillar structure P**2** may be a second channel structure. The first pillar structure P**1** may include a first channel layer **53**A, a first conductive pad **54**A, and a first insulating core **55**A. The second pillar structure P**2** may include a second channel layer **53**B, a second conductive pad **54**B, and a second insulating core **55**B. Alternatively, however, the first pillar structure P**1** may be a first electrode structure and the second pillar structure P**2** may be a second electrode structure. The first electrode structure may include a first electrode layer instead of the second channel layer **53**A. The second electrode structure may include a second electrode layer instead of the second channel layer **53**B.
(59) According to an embodiment, trenches T that pass through the stacked structure ST and the pillar structures P may be formed. The trenches T may extend in depth to completely pass through the pillar structures P and reach the base **50**. The trenches T may extend in the first direction I and pass through at least two pillar structures P. Subsequently, the cutting structures **56** may be formed in the trenches T. The cutting structures **56** may be provided to insulate the first pillar structure P**1** and the second pillar structure P**2** from each other, and may include an insulating material.
(60) Referring to FIGS. **7**A to **7**C, a first slit SL**1** may be formed through the stacked structure ST. The first slit SL**1** may extend in a direction crossing the cutting structure **56**. The first slit SL**1** may extend in the second direction II and the first slit SL**1** may be spaced apart from the cutting structures **56**. The first slit SL**1** may have a depth to expose the first material layers **51** and extend to the base **50**.
(61) Subsequently, the first material layers **51** may be replaced with third material layers **57**. For example, when the first material layers **51** are sacrificial layers and the second material layers **52** are insulating layers, the first material layers **51** may be replaced by conductive layers. After the first material layers **51** are selectively etched, the third material layers **57** may be formed in areas from which the first material layers **51** are etched. However, a memory layer may be formed before the third material layers **57** are formed. In another example, when the first material layers **51** are conductive layers and the second material layers **52** are insulating layers, the first material layers **51** may be silicided. As a result, the gate structure GST in which the third material layers **57** and the second material layers **52** are stacked alternately with each other may be formed. Subsequently, a first slit structure **58** may be formed in the first slit SL**1**.
(62) Referring to FIGS. **8**A to **8**C, a second slit SL**2** may be formed through the gate structure GST. The second slit structure SLS**2** may pass through the gate structure GST at a shallower depth than the first slit structure **58** or the cutting structure **56**. The second slit SL**2** may extend in a direction

crossing the cutting structures **56**, and may extend in the second direction II. In the plane defined by the first direction I and the second direction II, the second slit SL**2** may have a linear shape, a zigzag shape, a wave shape, or the like.
(63) The second slit SL**2** may be formed between the pillar structures P. When the second slit SL**2** is formed, the cutting structure **56** or the pillar structure P may be etched together with the stacked structure ST. Therefore, the cutting structure **56** or the pillar structure P may be exposed through the second slit SL**2**. The second slit SL**2** may be formed to cross the cutting structure **56**. One cutting structure **56** may be separated into a plurality of patterns by the second slit SL**2**.
(64) Subsequently, a second slit structure **59** may be formed in the second slit SL**2**. The second slit structure **59** may include an insulating material. At least one uppermost third material layer **57** may be separated into a plurality of patterns by the second slit structure **59**. The second slit structure **59** may contact a neighboring cutting structure **56** or pillar structure P.
(65) Though not shown in the drawings, interconnection lines coupled to the pillar structures P may be formed. According to an embodiment, at least one first bit line which extends in the first direction I and is coupled to the first pillar structures P**1** may be formed, and at least one second bit line which extends in the first direction I and is coupled to the second pillar structures P**2** may be formed.
(66) According to the above-described manufacturing method, one pillar structure P may be separated into the plurality of pillar structures P**1** and P**2** using the cutting structure **56**. Therefore, the number of memory cells realized with one pillar structure P may be increased. In addition, by forming the second slit SL**2** in a direction crossing the cutting structure **56**, the processes of replacing the first material layers **51** with the third material layers **57** may be improved.
(67) FIGS. **9**A and **9**B are diagrams illustrating a method of manufacturing a semiconductor device according to an embodiment of the present disclosure.
(68) FIGS. **9**A and **9**B are diagrams for visualizing the influence of the arrangements of cutting structures **56** and **56**′ and the first slit SL**1** during the processes of replacing the first material layers **51** with the third material layers **57**. To replace the first material layers **51** with the third material layers **57**, after the first material layers **51** are selectively etched, the third material layers **57** may be deposited onto areas from which the first material layers **51** are etched. An etch process of the first material layers **51** may be performed using a chemical such as an etchant. A chemical may be introduced between the pillar structures P and the cutting structures **56** and **56**′ (as indicated by arrows) to selectively etch the first material layers **51**.
(69) Referring to FIG. **9**A, the first slits SL**1** may be formed in parallel with the cutting structures **56**′. A flow path of the chemical may be restricted by the cutting structures **56**′ extending in the second direction II. Therefore, the chemical may be resisted from flowing between the cutting structures **56**′ neighboring each other in the first direction I, and a region R of the first material layers may remain without being etched.
(70) Referring to FIG. **9**B, the first slits SL**1** may be formed to cross the cutting structures **56**. According to an embodiment, the first slits SL**1** may be formed to be perpendicular to the cutting structures **56**. Because the cutting structures **56** extend in the first direction I, the flow path of the chemical may not be restricted, or may be less restricted. Therefore, the chemical may be sufficiently introduced between the cutting structures **56** and between the pillar structures P. As a result, a region where the first material layers **51** remain without being etched may be reduced.
(71) FIG. **10** is a diagram illustrating a memory system **1000** according to an embodiment of the present disclosure.
(72) Referring to FIG. **10**, the memory system **1000** may include a memory device **1200** configured to store data and a controller **1100** configured to perform communications between the memory device **1200** and a host **2000**.
(73) The host **2000** may be a device or system configured to store data in the memory system **1000** or retrieve data from the memory system **1000**. The host **2000** may generate requests for various

operations and output the generated requests to the memory system **1000**. The requests may include a program request for a program operation, a read request for a read operation, and an erase request for an erase operation. The host **2000** may communicate with the memory system **1000** by using at least one interface protocol among, for example, Peripheral Component Interconnect Express (PCIe), Advanced Technology Attachment (ATA), Serial ATA (SATA), Parallel ATA (PATA), Serial Attached SCSI (SAS), Non-Volatile Memory express (NVMe), Universal Serial Bus (USB), Mult Media Card (MMC), Enhanced Small Disk Interface (ESDI), and Integrated Drive Electronics (IDE).

(74) The host **2000** may include at least one of a computer, a portable digital device, a tablet, a digital camera, a digital audio player, a television, a wireless communication device, or a cellular phone. However, embodiments of the disclosed technology are not limited thereto.

(75) The controller **1100** may control overall operations of the memory system **1000**. The controller **1100** may control the memory device **1200** in response to the requests of the host **2000**. The controller **1100** may control the memory device **1200** to perform a program operation, a read operation and an erase operation at the request of the host **2000**. Alternatively, the controller **1100** may perform a background operation for performance improvement of the memory system **1000** in the absence of the request from the host **2000**.

(76) To control the operations of the memory device **1200**, the controller **1100** may transfer a control signal and a data signal to the memory device **1200**. The control signal and the data signal may be transferred to the memory device **1200** through different input/output lines. The data signal may include a command, an address, or data. The control signal may be used to differentiate periods in which the data signal is input.

(77) The memory device **1200** may perform a program operation, a read operation, and an erase operation in response to control of the controller **1100**, The memory device **1200** may include volatile memory that loses data when a power supply is blocked, or non-volatile memory that retains data in the absence of supplied power. The memory device **1200** may be a semiconductor device which has the structure as described above with reference to FIGS. **1**A to **1**D, FIGS. **2**A to **2**C, FIGS. **3**A to **3**C, and FIGS. **4**A to **4**D. The memory device **1200** may be a semiconductor device manufactured by the method as described above with reference to FIGS. **5**A and **5**B, FIGS. **6**A and **6**B, FIGS. **7**A to **7**C, and FIGS. **8**A to **8**C. According to an embodiment, a semiconductor device may include a gate structure that includes conductive layers and insulating layers stacked alternately with each other, channel structures that pass through the gate structure and are arranged in a first direction, a cutting structure that extends in the first direction and successively passes through the channel structures, and a first slit structure that passes through the gate structure and extends in a second direction crossing the first direction.

(78) FIG. **11** is a diagram illustrating a memory system **30000** according to an embodiment of the present disclosure.

(79) Referring to FIG. **11**, the memory system **30000** may be incorporated into a cellular phone, a smart phone, a tablet, a personal computer (PC), a personal digital assistant (PDA), or a wireless communication device. The memory system **30000** may include a memory device **2200** and a controller **2100** controlling the operations of the memory device **2200**.

(80) The controller **2100** may control a data access operation of the memory device **2200**, for example, a program operation, an erase operation, or a read operation of the memory device **2200** in response to control of a processor **3100**.

(81) The data programmed into the memory device **2200** may be output through a display **3200** in response to control of the controller **2100**.

(82) A radio transceiver **3300** may exchange a radio signal through an antenna ANT. For example, the radio transceiver **3300** may change the radio signal received through the antenna ANT into a signal which may be processed by the processor **3100**. Therefore, the processor **3100** may process the signal output from the radio transceiver **3300** and transfer the processed signal to the controller

2100 or the display **3200**. The controller **2100** may transfer the signal processed by the processor **3100** into the memory device **2200**, In addition, the radio transceiver **3300** may change a signal output from the processor **3100** into a radio signal and output the radio signal to an external device through the antenna ANT. A control signal for controlling the operations of the host or data to be processed by the processor **3100** may be input by an input device **3400**, and the input device **3400** may include a pointing device, such as a touch pad and a computer mouse, a keypad, or a keyboard. The processor **3100** may control the operations of the display **3200** so that data output from the controller **2100**, data output from the radio transceiver **3300**, or data output from an input device **3400** may be output through the display **3200**.
(83) According to an embodiment, the controller **2100** capable of controlling the operations of the memory device **2200** may be realized as a portion of the processor **3100**, or as a separate chip from the processor **3100**.
(84) FIG. **12** is a diagram illustrating a memory system **40000** according to an embodiment of the present disclosure.
(85) Referring to FIG. **12**, the memory system **40000** may be incorporated into a personal computer (PC), a tablet PC, a net-book, an e-reader, a personal digital assistant (PDA), a portable multimedia player (PMP), an MP3 player, or an MP4 player.
(86) The memory system **40000** may include the memory device **2200** and the controller **2100** that controls a data processing operation of the memory device **2200**.
(87) A processor **4100** may output data stored in the memory device **2200** through a display **4300** according to data input through an input device **4200**. Examples of the input device **4200** may include a pointing device such as a touch pad or a computer mouse, a keypad, or a keyboard.
(88) The processor **4100** may control overall operations of the memory system **40000** and control operations of the controller **2100**. According to an embodiment, the controller **2100** capable of controlling the operations of the memory device **2200** may be realized as a portion of the processor **4100**, or as a separate chip from the processor **4100**.
(89) FIG. **13** is a block diagram illustrating a memory system **50000** according to an embodiment of the present disclosure.
(90) Referring to FIG. **13**, the memory system **50000** may be incorporated into an image processor, for example, a digital camera, a cellular phone with a digital camera attached thereto, a smart phone with a digital camera attached thereto, or a table PC with a digital camera attached thereto.
(91) The memory system **50000** may include the memory device **2200** and the controller **2100** that controls a data processing operation of the memory device **2200**, for example, a program operation, an erase operation, or a read operation.
(92) An image sensor **5200** of the memory system **50000** may convert an optical image into digital signals. The converted digital signals may be transferred to a processor **5100** or the controller **2100**. In response to control of the processor **5100**, the converted digital signals may be output through a display **5300** or stored in the memory device **2200** through the controller **2100**. In addition, the data stored in the memory device **2200** may be output through the display **5300** in response to control of the processor **5100** or the controller **2100**.
(93) According to an embodiment, the controller **2100** capable of controlling the operations of the memory device **2200** may be formed as a part of the processor **5100**, or a separate chip from the processor **5100**.
(94) FIG. **14** is a diagram illustrating a memory system **70000** according to an embodiment of the present disclosure.
(95) Referring to FIG. **14**, the memory system **70000** may include a memory card or a smart card. The memory system **70000** may include the memory device **2200**, the controller **2100**, and a card interface **7100**.
(96) The controller **2100** may control the data exchange between the memory device **2200** and the card interface **7100**. According to an embodiment, the card interface **7100** may be, but is not

limited to, a secure digital (SD) card interface or a multi-media card (MMC) interface.
(97) The card interface **7100** may interface the data exchange between a host **60000** and the controller **2100** according to a protocol of the host **60000**, According to an embodiment, the card interface **7100** may support a Universal Serial Bus (USB) protocol and an InterChip (IC)-USB protocol.
(98) The card interface **7100** may refer to hardware capable of supporting a protocol which is used by the host **60000**, software installed in the hardware, or a signal transmission method.
(99) When the memory system **70000** is connected to a host interface **6200** of the host **60000** such as a PC, a tablet PC, a digital camera, a digital audio player, a cellular phone, a console video game hardware, or a digital set-top box, the host interface **6200** may perform data communication with the memory device **2200** through the card interface **7100** and the controller **2100** in response to control of a microprocessor **6100**.
(100) By three-dimensionally stacking memory cells, the density of integration of a semiconductor device may be improved. In addition, a semiconductor device having a stabilized structure and improved reliability may be provided.

Claims

1. A semiconductor device, comprising: a gate structure including conductive layers and insulating layers alternately stacked with each other; channel structures passing through the gate structure in a vertical direction and arranged in a first direction, wherein the vertical direction is orthogonal to the first direction; a cutting structure passing through the channel structures in the first direction and passing through the channel structures in the vertical direction; and a first slit structure passing through the gate structure in the vertical direction and extending in a second direction, the second direction being orthogonal to the first direction and the vertical direction, wherein the first slit structure and the channel structures are physically spaced apart from each other.
2. The semiconductor device of claim 1, wherein each of the channel structures is separated into a first channel structure and a second channel structure by the cutting structure.
3. The semiconductor device of claim 2, further comprising: at least one first bit line extending in the first direction and coupled to first channel structures; and at least one second bit line extending in the first direction and coupled to second channel structures.
4. The semiconductor device of claim 1, further comprising a second slit structure penetrating into the gate structure at a shallower depth than both the first slit structure and the cutting structure, the second slit structure extending in the second direction.
5. The semiconductor device of claim 4, wherein the cutting structure and the second slit structure contact each other.
6. The semiconductor device of claim 1, wherein the cutting structure includes an insulating material.
7. A semiconductor device, comprising: a gate structure including conductive layers and insulating layers alternately stacked with each other; pillar structures passing through the gate structure in a vertical direction and arranged in a first direction, wherein the vertical direction is orthogonal to the first direction; a cutting structure passing through the pillar structures in the first direction, passing through the pillar structures in the vertical direction, and separating each of the pillar structures into a first pillar structure and a second pillar structure; a first slit structure passing through the gate structure in the vertical direction and extending in a second direction, wherein the second direction is orthogonal to the first direction and the vertical direction; a first interconnection line extending in the first direction, the first interconnection line coupled to the first pillar structures; and a second interconnection line extending in the first direction, the second interconnection line coupled to the second pillar structures, wherein the first slit structure and the pillar structures are physically spaced apart from each other.

8. The semiconductor device of claim 7, further comprising: first contact plugs coupled to the first pillar structures, respectively, the first contact plugs coupling the first pillar structures to the first interconnection line; and second contact plugs coupled to the second pillar structures, respectively, the second contact plugs coupling the second pillar structures to the second interconnection line.
9. A method of manufacturing a semiconductor device, the method comprising: forming a stacked structure; forming channel structures passing through the stacked structure in a vertical direction and arranged in a first direction, wherein the vertical direction is orthogonal to the first direction; forming a cutting structure passing through the channel structures in the first direction and passing through the channel structures in the vertical direction; and forming a first slit structure passing through the stacked structure in the vertical direction and extending in a second direction, the second direction being orthogonal to the first direction and the vertical direction, wherein the first slit structure and the channel structures are physically spaced apart from each other.
10. The method of claim 9, wherein forming the cutting structure comprises etching the channel structures so that each of the channel structures is separated into a first channel structure and a second channel structure.
11. The method of claim 10, further comprising: forming at least one first bit line extending in the first direction and coupled to first channel structures; and forming at least one second bit line extending in the first direction and coupled to second channel structures.
12. The method of claim 9, wherein forming the cutting structure comprises: forming a trench crossing at least two channel structures; and forming the cutting structure to include an insulating material in the trench.
13. The method of claim 9, wherein forming of the first slit structure comprises: forming a first slit passing through the stacked structure including alternately stacked first material layers and second material layers, the first slit extending in the second direction; replacing the first material layers with third material layers through the first slit; and forming the first slit structure in the first slit.
14. The method of claim 9, further comprising forming a second slit structure penetrating into the stacked structure at a shallower depth than the first slit structure and extending in the second direction.
15. The method of claim 14, wherein forming the second slit structure comprises: forming a second slit by etching the stacked structure and the cutting structure; and forming the second slit structure in the second slit.